Midterm

Algorithms

[1] The worst case time complexity of insertion sort is

(a) O(nlogn) (b) O(logn) (c) O(n) (d) $O(n^2)$

[2] The number of errors in the following is

```
PARTITION (A, p, r)
x = A[r]
I = p –1
for j = p to r –1
    if A[ j ] ≤ x
        i = i + 1
        exchange A[i] with A[j]
exchange A[i + 1] with A[r]
return i + 1
```

(a) 1 (b) no correct answer (c) 4 (d) 3

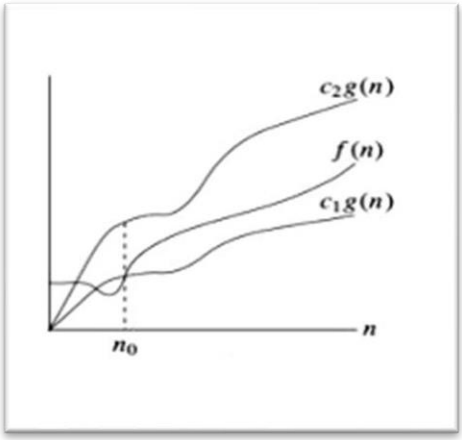
[3] The following is related to

```
for k = p to r
    if L[i] ≤ R[j]
        A[k] = L[i]
        i = i + 1
    else A[k] = R[j]
        j = j + 1
```

(a) Insertion sort (b) Quick sort (c) Heap sort (d) Merge sort

[4] The f(n) equals

(a) f(n) = Ω(g(n))
(b) f(n) = Θ(g(n))
(c) f(n) = O(g(n))
(d) no correct answer



[5] Based on the merge sort, the value of r in the subsequent array at the start is

36	25	40	2	7	80	15

(a) 36 (b) 80 (c) 2 (d) 15

[6] Based on the merge sort, the value of p in the subsequent array at the start is

36	25	40	2	7	80	15

(a) 36 (b) 80 (c) 2 (d) 15

[7] Based on the merge sort, the value of r in the subsequent array at the end of sorting is

36	25	40	2	7	80	15

(a) 36 (b) 80 (c) 2 (d) 15

[8] Based on the merge sort, the value of p in the subsequent array at the end of sorting is

36	25	40	2	7	80	15

(a) 36 (b) 80 (c) 2 (d) 15

[9] How many passes does an insertion sort algorithm consist of for **N** numbers?

(a) N (b) N^2 (c) $N-1$ (d) $N+1$

[10] In Merge sort, divide the n-element sequence to be sorted into two subsequences of ..... elements each

(a) n (b) n-2 (c) n+2 (d) n/2

[11] What is the worst case time complexity of a quick sort algorithm?

(a) O(nlogn) (b) O(logn) (c) O(n) (d) $O(n^2)$

[12] Based on the insertion sort, the time complexity of the following array is

10	20	30	40	50	60

(a) O(nlogn) (b) O(logn) (c) O(n) (d) O(n^2)

[13] What will be the number of passes to sort the elements using insertion sort?

10	20	30	40	50	60

(a) 6 (b) no correct answer (c) 7 (d) 5

[14] How many keys will be needed to sort the elements using insertion sort?

10	20	30	40	50	60

(a) 6 (b) no correct answer (c) 7 (d) 5

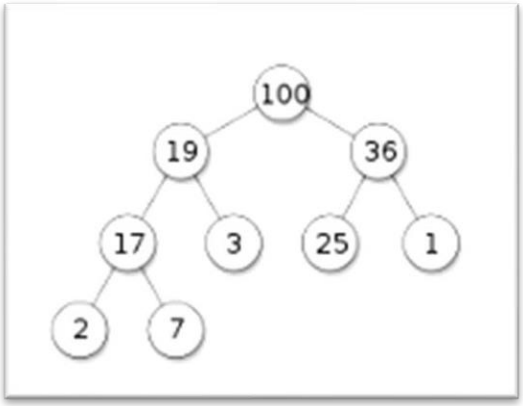
[15] The following tree achieves the

(a) max heap

(b) no correct answer

(c) heap

(d) min heap



[16] Based on the merge sort, the total running time is

(a) O(nlogn) (b) O(logn) (c) O(n) (d) O(n^2)

[17] Insertion sorting an array of random values takes

(a) O(nlogn) (b) O(logn) (c) O(n) (d) O(n^2)

[18] The running time of a heap sort algorithm is

(a) O(N) (b) O(log N) (c) O(N^2) (d) O(N log N)

[19] The number of errors in the following is

```
PARENT(i)
1  return ⌊i/2⌋

LEFT(i)
1  return 2i

RIGHT(i)
1  return 2i + 1
```

(a) 1 (b) no correct answer

(c) 4 (d) 3

[20] The number of errors in the following is

```
MERGE-SORT(A, p, r)
1  if p + r
2      q = ⌊(p + r)/2⌋
3      MERGESORT(A, p, q)
4      MERGESORT(A, q + 1, r)
5      MERGE(A, p, q, r)
```

(a) 1 (b) no correct answer (c) 4 (d) 3